

# **Data Visualization and Chart Types**

Structured Lecture Notes

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December 27, 2025

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# 1 Overview of Charts and Visualizations

Data visualization plays a crucial role in transforming raw data into meaningful insights. Charts and visual representations allow us to identify patterns, trends, comparisons, and relationships that may not be immediately apparent from tables or raw numbers. In this section, we explore several commonly used chart types and explain how each can be used effectively to tell a data-driven story.

## 1.1 Line Charts

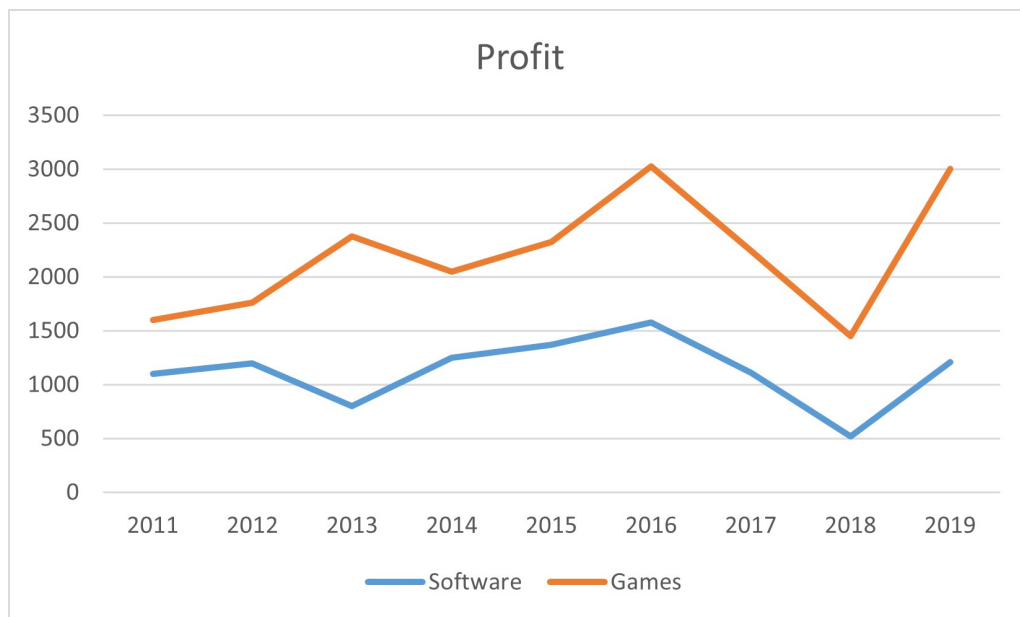


Figure 1: Example of Line Charts.

A **line chart** is one of the most effective tools for visualizing trends over a continuous variable. It is especially useful when the goal is to understand how data changes over time.

### Key Characteristics:

- The horizontal axis typically represents a continuous variable such as time.
- The vertical axis represents the measured value (e.g., sales, temperature, revenue).
- Data points are connected using straight lines to highlight trends.

**Use Case Example:** If time is the continuous variable, a line chart can show how the sales of one or multiple products increase or decrease over several months or years. This makes it easy to observe growth patterns, seasonal effects, or sudden changes.

## 1.2 Pie Charts

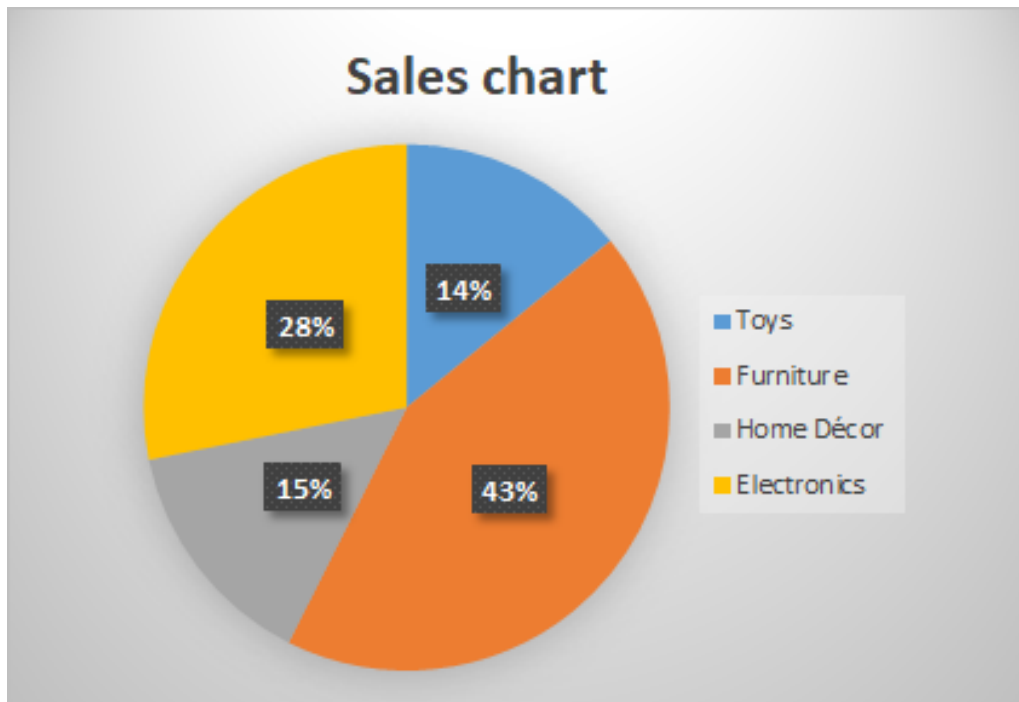


Figure 2: Example of Pie Charts.

A **pie chart** is used to represent the proportional breakdown of a whole into its individual components.

### **Key Characteristics:**

- Each slice represents a category or sub-part.
- The total of all slices equals 100 percent.
- Best suited for displaying static data rather than trends.

**Use Case Example:** In a marketing campaign with categories such as Social Sites, Native Advertising, Paid Influencers, and Live Events, a pie chart can quickly show the proportion of leads generated by each category.

## 1.3 Bar Charts

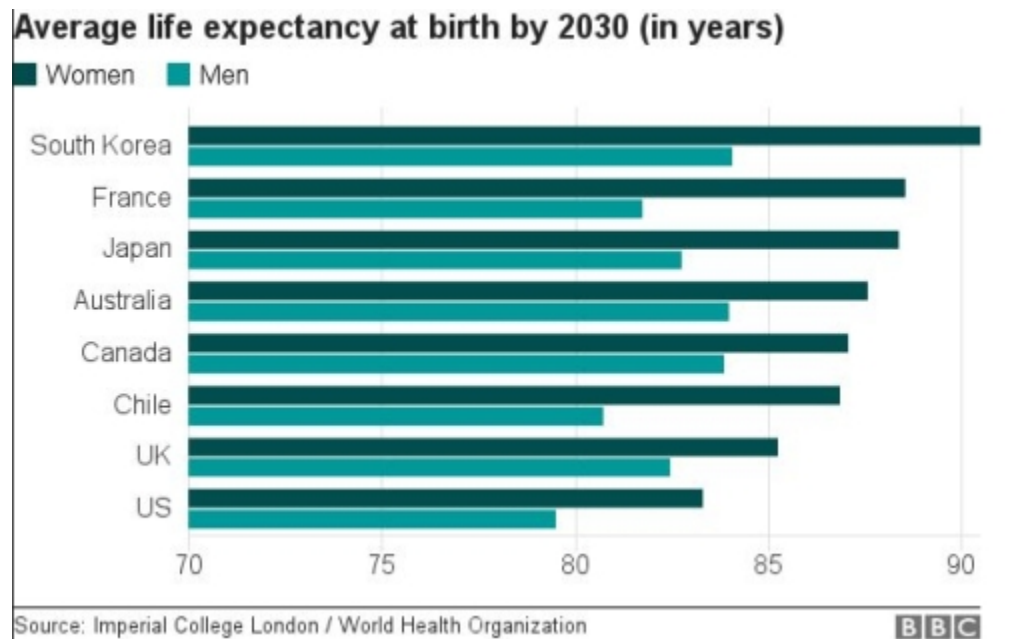


Figure 3: Example of bar Charts.

The **bar chart** is one of the most commonly used chart types due to its simplicity and clarity.

### Key Characteristics:

- Uses rectangular bars to represent data values.
- Ideal for comparing different categories or parts of a whole.
- Can be displayed horizontally.

**Use Case Example:** A bar chart can be used to compare the population sizes of ten different countries, making relative differences easy to interpret.

## 1.4 Stacked Bar Charts

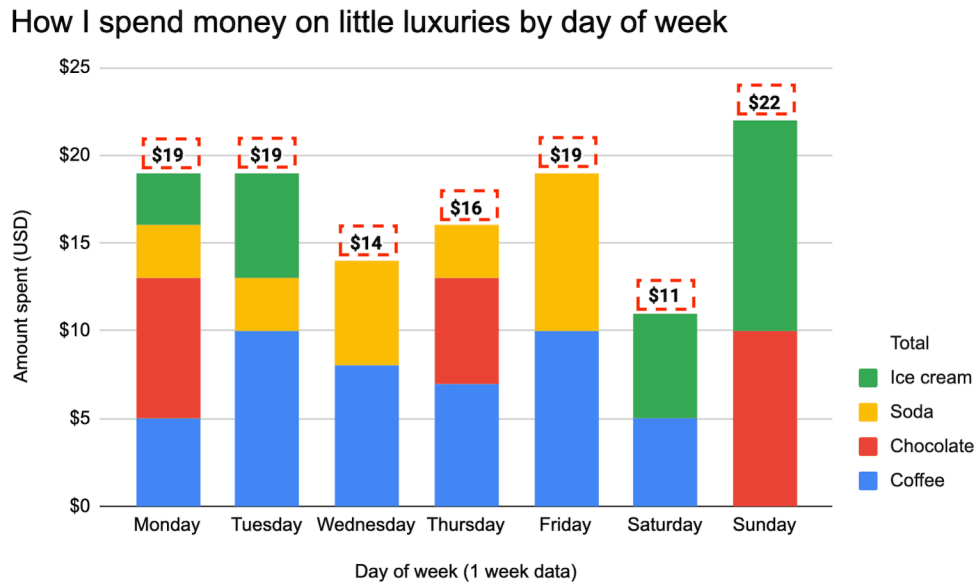


Figure 4: Example of Stacked bar Charts.

A **stacked bar chart** extends the basic bar chart by dividing each bar into multiple sub-bars.

### Key Characteristics:

- Each bar represents a total value.
- Sub-bars show the contribution of different components.
- Useful for comparing both totals and sub-category distributions.

**Use Case Example:** Displaying country populations divided into age groups (e.g., children, youth, adults, elderly) allows for deeper demographic analysis.

## 1.5 Column Charts

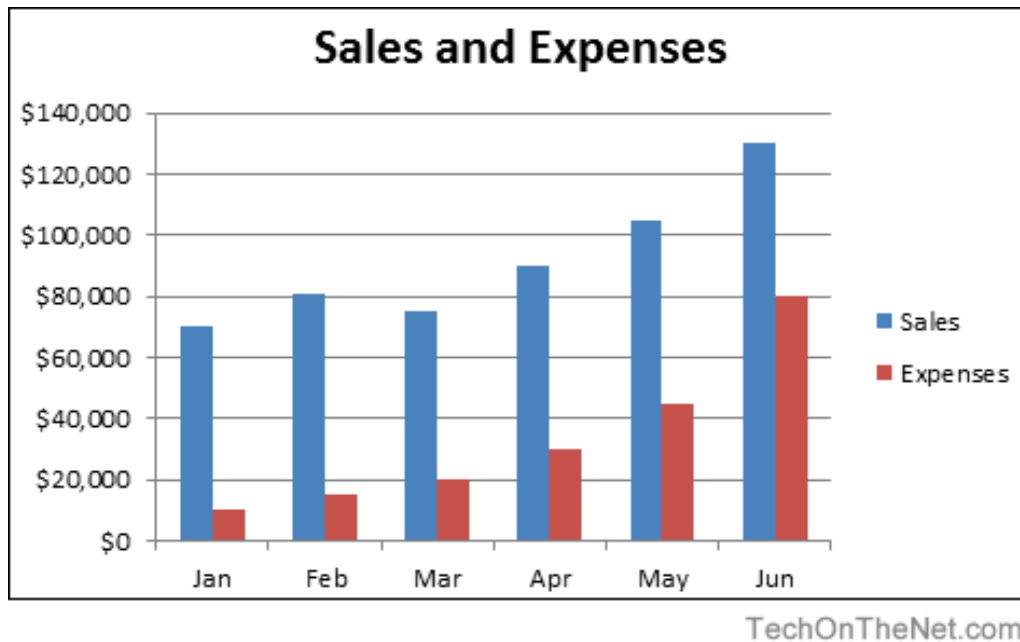


Figure 5: Example of Column Charts.

A **column chart** is similar to a bar chart but is oriented vertically.

### Key Characteristics:

- Best suited for showing changes over time.
- Useful for side-by-side comparisons.
- Handles positive and negative values effectively.

**Use Case Example:** Comparing monthly page views and user session times on a website highlights performance trends across time periods.

## 1.6 Treemaps

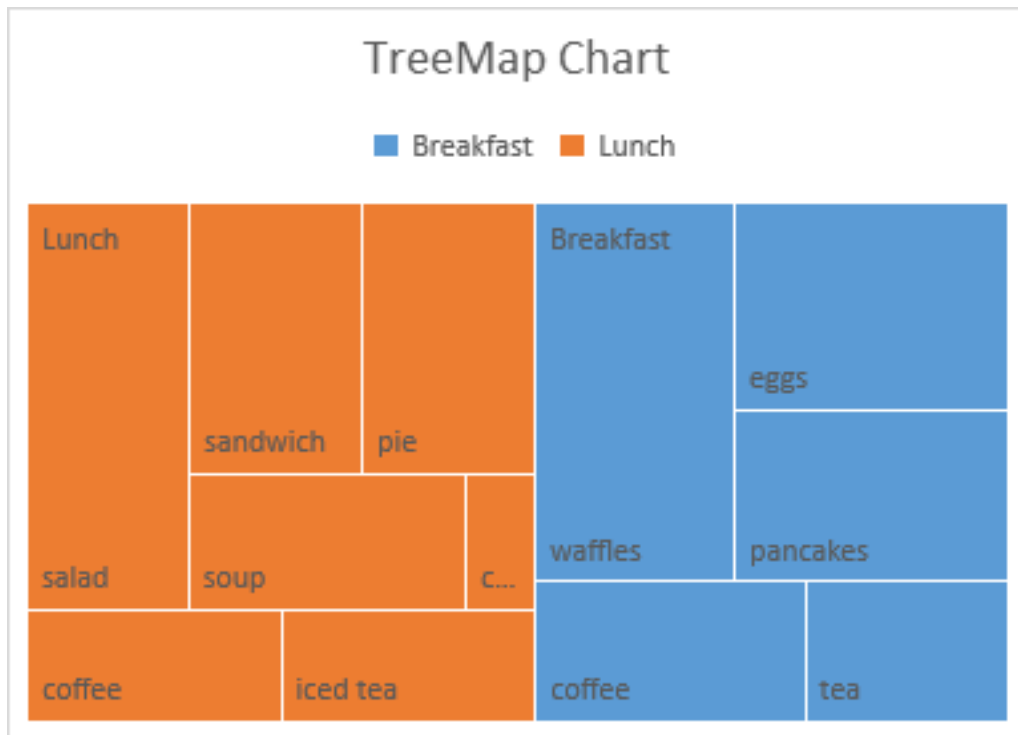


Figure 6: Example of Treemap Charts.

A **treemap** displays hierarchical data using nested rectangles.

### Key Characteristics:

- Rectangle size represents magnitude (e.g., population).
- Color represents a secondary variable (e.g., employment rate).
- Enables drill-down into subcategories.

**Use Case Example:** Visualizing statewide employment data within a country allows users to compare regions and explore sub-regions interactively.



## 1.7 Funnel Charts

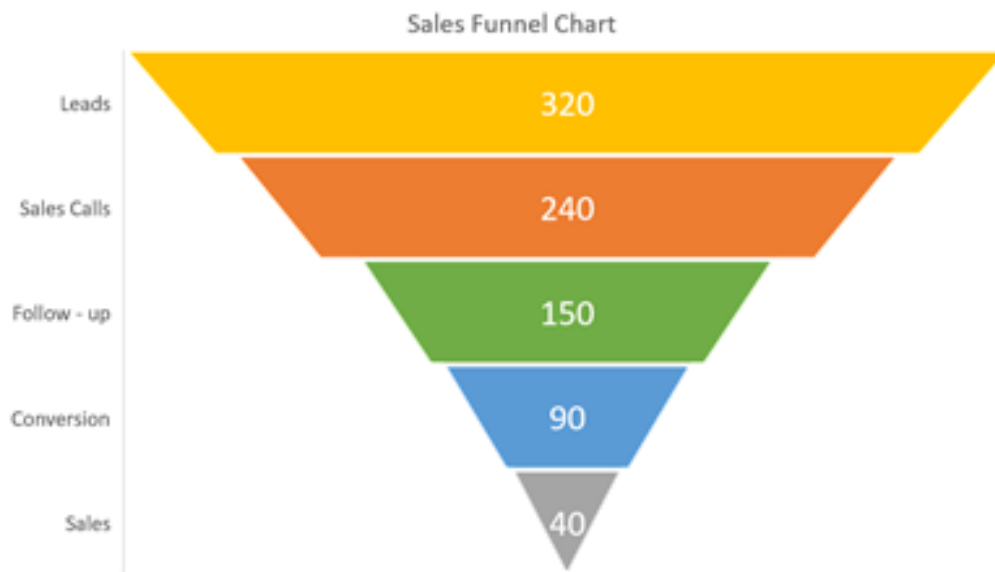


Figure 7: Example of Funnel Charts.

A **funnel chart** is designed to show stages in a sequential process.

**Key Characteristics:**

- Each stage represents a step in a process.
- The width decreases to reflect drop-offs.
- Ideal for conversion analysis.

**Use Case Example:** Tracking sales conversion from lead generation to final purchase helps identify where potential customers are lost.

## 1.8 Scatter Charts

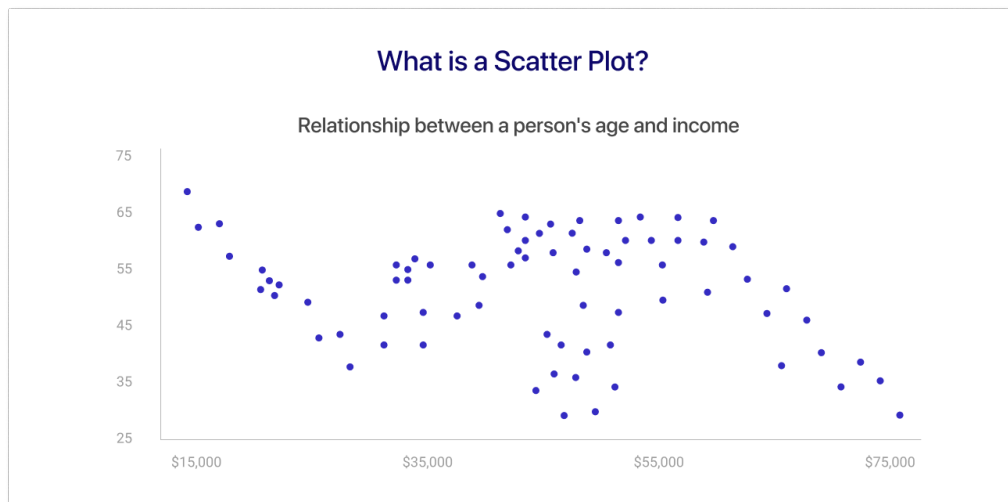


Figure 8: Example of Scatter Charts.

A **scatter chart** plots individual data points to reveal relationships between variables.

**Key Characteristics:**

- Each point represents a data observation.
- Color and size can encode additional variables.
- Useful for identifying correlations and clusters.

**Use Case Example:** Analyzing product lines by units sold and revenue helps identify high-performing and underperforming products.

## 1.9 Bubble Charts

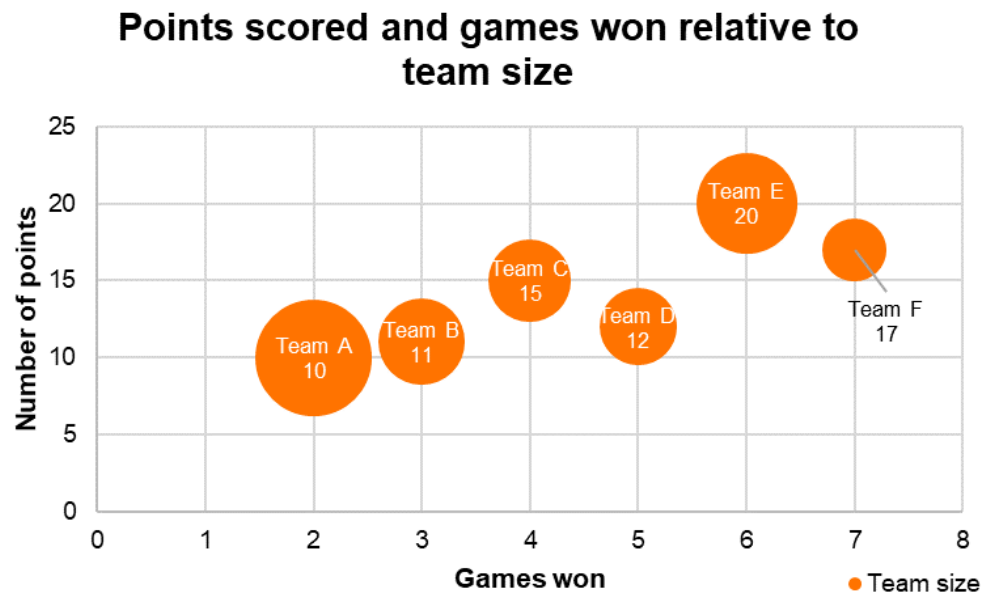


Figure 9: Example of Bubble Charts.

A **bubble chart** is a variation of the scatter chart with an added dimension.

**Key Characteristics:**

- Bubble size represents magnitude.
- Effective for comparing relative importance.

**Use Case Example:** Understanding areas of significant expenditure in an organization's sales budget becomes more intuitive.

## 1.10 Sparklines

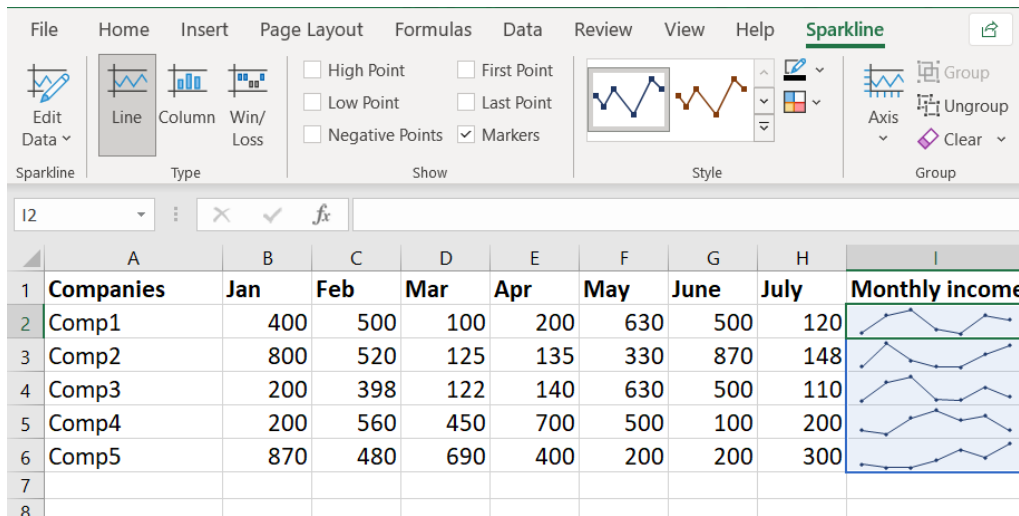


Figure 10: Example of Sparkling Charts.

**Sparklines** are small, simple visualizations without axes.

### Key Characteristics:

- No labels or coordinates.
- Focus on overall trend rather than exact values.
- Ideal for dashboards and summaries.

**Use Case Example:** Displaying stock price movement from market open to close provides a quick trend overview.

## 1.11 Summary

Charts and visualizations shape data into meaningful insights by emphasizing trends, proportions, and relationships. Selecting the appropriate chart type is essential for effective communication. In subsequent sections, we will explore how to create and configure these charts practically using Excel.

## 2 Additional Chart Types

In this section, we discuss several additional chart types that are commonly used for analyzing complex datasets, relationships between variables, and data distributions.

## 2.1 Treemaps

Treemaps are a powerful visualization technique used to compare values across different levels of a hierarchy.

- Treemaps compare values across hierarchy levels.
- They show proportions using rectangles of varying sizes.
- A treemap is a good way to display lots of data in one graphic.
- Color, size, and closeness of shapes represent hierarchical data categories.

Treemaps are especially useful when dealing with large, hierarchical datasets where both relative size and categorical relationships need to be communicated clearly.

## 2.2 Scatter Charts

Scatter charts are used to compare two sets of numerical data values and to analyze the relationship between them.

- Scatter charts compare two sets of numerical data values.
- They show relationships between data points.
- They combine two sets of 'x' and 'y' axis values into single data points.
- Data points are often displayed in clusters.
- Scatter charts are commonly used to compare statistical, scientific, and engineering data values.

Scatter charts are particularly effective for identifying trends, correlations, patterns, and outliers within datasets.

## 2.3 Histograms

Histograms are used to visualize the distribution of numerical data by grouping values into ranges known as bins.

- Histograms show the distribution of data grouped into 'bins'.
- They look similar to bar charts but are fundamentally different.
- Bar charts compare individual data values, while histograms display data distribution.

Histograms help analysts understand the spread, central tendency, and shape of a dataset, making them essential for statistical analysis.

## 3 Advanced Charts in Excel

In this section, we explore more advanced chart types available in Excel. The focus is on filled map charts, sparklines, and a brief overview of several other specialized charts that are useful for advanced data analysis and visualization.

### 3.1 Filled Map Charts

A **filled map chart** is a type of chart used to compare values and show categories across geographical regions.

- Filled map charts are suitable for data containing geographical regions such as countries, states, or postal codes.
- They visually represent values using color intensity across different regions.
- Darker colors typically indicate higher values, while lighter colors indicate lower values.

To create a filled map chart, data must first be organized correctly. In the car sales workbook, data containing the country of sale and the sum of unit sales is copied from a pivot table and pasted beside it. After selecting this data, the Filled Map chart option is chosen from the Map category in the Charts group.

Once created, the chart displays the sum of unit sales of cars across different countries. The chart title can be edited directly by double-clicking the title textbox and renaming it, for example, to “*Sum of Unit Sales of Cars by Country*”. Chart styles can then be applied to customize the visual appearance.

From the filled map visualization:

- Dark blue regions indicate countries with the highest unit sales, such as the United States.
- Medium blue shades represent moderate sales, covering areas like Canada, Western Europe, and Scandinavia.
- Very light or almost white shades indicate the lowest sales, seen in regions such as Eastern Europe, India, Japan, and Australia.

This chart type is particularly effective for identifying geographical patterns and regional performance differences.

## 3.2 Sparklines

Sparklines are small, compact charts placed inside individual worksheet cells to represent trends in data.

- Sparklines are used to show data trends such as seasonal fluctuations, economic cycles, and changes in share prices or rates.
- They can also highlight maximum and minimum values within a dataset.
- Sparklines provide the greatest impact when placed close to the data they represent.

To create sparklines, data from adjacent columns such as *Unit Sales Q1*, *Unit Sales Q2*, *Unit Sales Q3*, and *Unit Sales Q4* is selected. A line sparkline is then chosen from the Sparklines group.

The location for the sparkline is specified by either typing a cell reference or directly selecting a cell in the worksheet. Excel automatically applies absolute references using dollar symbols. Once created, the sparkline can be copied down the column to apply it to other rows.

After creation:

- The column containing sparklines is labeled as *Quarter Sales Trends*.
- Column width and row height are adjusted for better visibility.
- Maximum and minimum values are displayed on the sparklines.
- Chart styles and line weights are adjusted to improve clarity.

From the sparklines, trends can be observed clearly. For example, Ford Escort unit sales start low in the first quarter, increase during quarters two and three, and then decline in the fourth quarter. Overall analysis shows that quarter three generally has the highest unit sales across most Ford models, with some exceptions such as Mustangs peaking in quarter four and Focus models in quarter two.

## 3.3 Other Available Chart Types in Excel

Excel also provides several other specialized chart types for specific analytical needs.

- **Waterfall Charts:** Used to show the cumulative effect of a series of positive and negative values. These are commonly applied to financial data involving inflows and outflows.
- **Funnel Charts:** Used to display progressively smaller stages in a process, making them suitable for conversion or pipeline analysis.

- **Stock Charts:** Used to show trends in stock performance over time, typically including values such as volume, open, high, low, and close prices.
- **Surface Charts:** Used to show trends across two dimensions, displayed as 3-D surface areas or 2-D contour charts. These are most effective when both categories and data series are numeric.
- **Radar Charts:** Used to show values relative to a central point and are best suited when categories are not directly comparable.

### 3.4 Summary

In this section, we learned how to create and interpret filled map charts and sparklines in Excel, and we reviewed several other advanced chart types. These charts enable deeper insights by highlighting geographical patterns, trends over time, cumulative effects, and relative comparisons across multiple dimensions.